

Covariance & Correlation

Simple Data Science Notes • Concepts, examples, formulas and feature selection

1. What Are Covariance and Correlation?

Both covariance and correlation tell us how two variables move in relation to each other.

Example: Temperature and ice cream sales. If temperature increases and ice cream sales also increase, the variables have a positive relationship.

2. Covariance

Simple definition: Covariance tells us whether two variables tend to increase or decrease together.

Positive covariance: X increases while Y tends to increase, or both decrease together.

Negative covariance: X increases while Y tends to decrease.

Near-zero covariance: There is little or no linear relationship.

Situation	Example	Covariance
Positive	Temperature $\uparrow$ $\rightarrow$ Ice cream sales $\uparrow$	> 0
Negative	Price $\uparrow$ $\rightarrow$ Sales $\downarrow$	< 0
Near zero	No clear linear pattern	≈ 0

3. Covariance Formula

$$\text{Cov}(X, Y) = \frac{\sum[(X_i - \bar{X})(Y_i - \bar{Y})]}{n}$$

X_i = individual X value $\bar{X}$ = mean of X Y_i = individual Y value $\bar{Y}$ = mean of Y

The important part is $(X_i - \bar{X})(Y_i - \bar{Y})$. It means: how far X is from its mean $\times$ how far Y is from its mean.

4. Why Does Covariance Show the Relationship?

Suppose $\bar{X} = 30$ and $\bar{Y} = 200$. If $X = 35$ and $Y = 250$:

$$(35 - 30)(250 - 200) = 5 \times 50 = 250 \rightarrow \text{positive contribution}$$

If $X = 35$ and $Y = 150$: $(35 - 30)(150 - 200) = 5 \times (-50) = -250 \rightarrow \text{negative contribution.}$

5. Important: Covariance Does NOT Prove Causation

Relationship $\neq$ Causation. If two variables move together, covariance only describes their co-movement. It does not prove that one variable causes the other.

6. Problem With Covariance

Covariance can range from $-\infty$ to $+\infty$ and depends on the units and scale of the variables. Therefore, covariance values are difficult to compare directly.

7. Correlation

Correlation is the standardized/scaled version of covariance. It measures both the direction and strength of a linear relationship.

$$-1 \leq \text{Correlation} \leq +1$$

Correlation	Meaning
+1	Perfect positive correlation
+0.8	Strong positive relationship
+0.5	Moderate positive relationship
0	No linear correlation
-0.5	Moderate negative relationship
-0.8	Strong negative relationship
-1	Perfect negative correlation

8. Correlation Formula

$$\text{Corr}(X,Y) = \frac{\sum[(X_i - \bar{X})(Y_i - \bar{Y})]}{\sqrt{[\sum(X_i - \bar{X})^2 \times \sum(Y_i - \bar{Y})^2]}}$$

The easiest form to remember for Data Science is:

$$\text{Correlation} = \text{Covariance} / (\text{SD}_X \times \text{SD}_Y)$$

9. Covariance vs Correlation

Covariance	Correlation
Measures direction of relationship	Measures direction + strength
Range: $-\infty$ to $+\infty$	Range: -1 to $+1$
Depends on units/scales	Standardized
Harder to compare	Easier to interpret

Memory trick: Covariance = direction. Correlation = direction + strength.

10. Real-World Example: Hours Studied vs Exam Marks

Hours Studied	Marks
1	40
2	50
3	60
4	70
5	80

As study hours increase, marks increase. Therefore covariance is positive, and if the relationship is nearly linear, correlation will be close to +1.

11. Correlation in Feature Selection

Suppose we are predicting house prices. We can calculate correlation between each feature and the target:

Feature	Correlation with Price	Interpretation
House Size	+0.85	Strong positive relationship
Bedrooms	+0.65	Moderate positive relationship
House Age	-0.45	Moderate negative relationship
Distance	-0.50	Moderate negative relationship
Random ID	+0.01	Almost no linear relationship

12. Highly Correlated Features

If two independent features are highly correlated, they may contain almost the same information. For example, house size in square feet and house size in square meters are nearly redundant.

Feature 1 ↔ Feature 2: very high correlation → possible redundancy → consider removing one.

This issue is related to **multicollinearity**, especially in models such as linear regression.

13. Important Limitation

Pearson correlation primarily measures **linear relationships**. A low correlation does not always mean there is no relationship.

For example, $Y = X^2$ can have a strong non-linear relationship even when ordinary correlation is close to zero in a symmetric dataset.

■ Final Memory Notes

Covariance: Which direction do the variables move?

Correlation: Which direction + how strongly?

Correlation range: -1 to +1

Feature vs Target: Low correlation → may consider removing the feature.

Feature vs Feature: Very high correlation → one may be redundant.

Warning: Correlation does NOT mean causation.